COMPUTER SCIENCE – FORM ONE

Teaching Notes: Computer System Handling and Care

Based on PDF Pages 102–113

Purpose: Simple teacher explanations, real-life examples, important concepts and classroom questions for guiding Form One students.

PAGE 102 – EFFECTS OF LIQUID ON A COMPUTER

Liquid can cause a short circuit

Book concept / teacher explanation: A computer uses electricity. Water can conduct electricity. When water reaches electronic circuits, it may allow electricity to flow through an unintended path and damage components.

Simple real-life example: Water is accidentally spilled on a laptop keyboard and reaches the electronic parts. The laptop may stop working.

Key point: Liquid + electricity can damage computer hardware.

Liquid can cause corrosion

Book concept / teacher explanation: Liquid can damage metal parts inside a computer over time. Corrosion can make electrical connections weak.

Simple real-life example: An iron gate can rust when exposed to water for a long time. Computer metal contacts can also be damaged by moisture.

Key point: Liquid damage may appear immediately or later.

Liquid can damage storage devices

Book concept / teacher explanation: Liquid can damage storage devices such as hard drives and may cause loss of access to stored files.

Simple real-life example: If a student's only copy of an assignment is on a damaged laptop, the assignment may be lost.

Class question: What can happen if water enters a computer containing important school records?

PAGE 102–103 – WHAT TO DO AFTER A LIQUID SPILL

Step 1: Turn off the computer

Book concept / teacher explanation: Turn the computer off immediately and disconnect it from electricity. Do not continue using a wet computer.

Simple real-life example: Example: If tea spills on a laptop, do not continue typing. Switch it off and disconnect it safely.

Key point: Never continue operating a wet computer.

Step 2: Disconnect external devices

Book concept / teacher explanation: Disconnect the keyboard, mouse, USB devices and other peripherals where it is safe to do so.

Simple real-life example: Example: Remove a wet external keyboard from the computer after the power has been disconnected.

Step 3: Dry a small spill carefully

Book concept / teacher explanation: Use a soft absorbent cloth to remove liquid from accessible external surfaces. Avoid spreading the liquid.

Simple real-life example: Example: Carefully wipe a few drops from the outside of a keyboard.

Step 4: Seek professional help

Book concept / teacher explanation: For a serious spill, a qualified technician should inspect the equipment. Do not attempt repairs without the required knowledge.

Simple real-life example: Example: If a large amount of water enters a desktop computer, disconnect it safely and seek technical assistance.

PAGE 103 – PREVENTION OF LIQUID DAMAGE

Keep drinks away from computers

Book concept / teacher explanation: Do not place water, juice, tea or other drinks next to a computer.

Simple real-life example: Example: Keep a bottle of water away from the laptop rather than placing it beside the keyboard.

Key point: Prevention is better than repairing liquid damage.

Use a safe working environment

Book concept / teacher explanation: Keep the computer in a clean, dry and organised place.

Simple real-life example: Example: Do not carry an open cup over a computer or put a cup on top of a computer case.

PAGE 103–104 – POWER MANAGEMENT

Power management

Book concept / teacher explanation: A computer needs a suitable and stable electricity supply. Very low or very high voltage can damage computer components.

Simple real-life example: Example: Electricity goes off, then returns suddenly. A power disturbance may affect the computer.

Key point: Unstable electricity can cause hardware damage and data loss.

Brownouts

Book concept / teacher explanation: A brownout is a temporary reduction in electrical voltage. It may be noticed when lights become dim or electrical equipment behaves abnormally.

Simple real-life example: A fan becomes slower and lights become dim because the supplied voltage has fallen.

Easy memory: Brownout = voltage goes DOWN.

Spikes

Book concept / teacher explanation: A spike is a sudden increase in voltage that lasts for a very short time. The book relates spikes to events such as lightning or power returning after a blackout.

Simple real-life example: Easy memory: Spike = sudden voltage increase.

Surges

Book concept / teacher explanation: A power surge is a sudden significant increase in electrical voltage. It can damage electronic equipment.

Simple real-life example: Example: An electrical disturbance during a storm can damage sensitive computer components.

Key point: Power fluctuations can damage computers.

PAGE 104–107 – PROTECTING COMPUTERS FROM POWER PROBLEMS

UPS – Uninterruptible Power Supply

Book concept / teacher explanation: A UPS provides temporary backup power when the main electricity supply fails. It gives the user time to save work and shut down safely.

Simple real-life example: Example: Power fails while a student is typing an assignment. The UPS keeps the computer running temporarily so the student can save the work and shut down.

Easy memory: UPS = Backup power.

Surge protector

Book concept / teacher explanation: A surge protector helps protect electrical equipment from sudden voltage spikes and surges.

Simple real-life example: Easy memory: Surge protector = protection against surges/spikes.

Voltage stabiliser

Book concept / teacher explanation: A voltage stabiliser helps maintain a more stable voltage supplied to equipment when the incoming voltage changes.

Simple real-life example: Easy memory: Stabiliser = stable voltage.

Important difference

Book concept / teacher explanation: UPS provides temporary backup power; a surge protector helps protect against surges/spikes; a voltage stabiliser helps maintain stable voltage.

Simple real-life example: Student memory: UPS = Backup | Surge protector = Surge | Stabiliser = Stable voltage.

PAGE 107 – RELOCATING A COMPUTER

Moving a computer safely

Book concept / teacher explanation: Shut down the computer, disconnect cables and carry it carefully using suitable packaging or a carrying case.

Simple real-life example: Example: Before moving a desktop, shut it down and disconnect the monitor, keyboard, mouse, power and network cables.

Laptop handling

Book concept / teacher explanation: Do not carry a laptop by its screen. Hold it securely from the base.

Simple real-life example: Example: Carry the laptop using both hands under the base rather than holding the display.

PAGE 108 – COMPUTER SOFTWARE CARE

Viruses and malicious software

Book concept / teacher explanation: Malicious software can enter computers through infected files, removable storage devices and unsafe downloads.

Simple real-life example: Example: An infected USB drive can transfer harmful software to a school computer.

Antivirus

Book concept / teacher explanation: Antivirus software helps detect and protect against harmful software.

Simple real-life example: Good practice: Scan USB drives, scan downloaded files and keep antivirus software updated.

Software updates

Book concept / teacher explanation: Operating systems, drivers and applications should be updated because updates can provide security patches, bug fixes and improved compatibility.

Simple real-life example: Example: Install genuine security updates rather than ignoring important update notifications.

PAGE 109 – PROPER COMPUTER SHUTDOWN

Step 1: Save your work

Book concept / teacher explanation: Save documents before shutting down.

Simple real-life example: Example: Save a file such as Computer Science Assignment before closing the computer.

Step 2: Close applications

Book concept / teacher explanation: Close Word, the browser, Excel and other applications before shutting down.

Step 3: Log out where appropriate

Book concept / teacher explanation: Log out of a user account when required, especially on shared computers.

Step 4: Use the Shutdown option

Book concept / teacher explanation: On Windows, use Start → Power → Shut down.

Step 5: Wait

Book concept / teacher explanation: Allow the computer to complete the shutdown process before disconnecting power.

Simple real-life example: Easy rule: Save → Close → Shut down → Wait.

Avoid forceful shutdown

Book concept / teacher explanation: Do not normally pull out the power cable or force the computer off while it is running. This can contribute to data loss, file corruption and system errors.

Simple real-life example: Example: Removing power while an assignment is being saved may cause the latest changes to be lost.

PAGE 110–111 – HANDLING PERIPHERALS AND CABLES

Keyboard care

Book concept / teacher explanation: Type gently. Do not hit the keys or place heavy objects on the keyboard.

Simple real-life example: Example: Do not put books or a heavy bag on top of a keyboard.

Mouse care

Book concept / teacher explanation: Handle the mouse gently and avoid dropping or throwing it.

Simple real-life example: Example: Move the mouse on a suitable surface and press its buttons normally.

Cable care

Book concept / teacher explanation: Do not excessively bend cables, pull them by the wire or place them near sharp objects.

Simple real-life example: Correct method: Hold the connector when disconnecting a cable.

Cable organisation

Book concept / teacher explanation: Use cable ties, clips or sleeves to keep cables organised and reduce tripping and accidental pulling.

Simple real-life example: Example: A cable crossing a classroom walkway can cause someone to trip and damage the equipment.

Inspect cables

Book concept / teacher explanation: Check for cuts, broken insulation, exposed wires and loose connectors. Replace damaged cables.

Simple real-life example: Safety point: Never use a cable with exposed electrical wires.

PAGE 111–112 – PASSWORD SECURITY

Meaning of a password

Book concept / teacher explanation: A password is a secret combination used to control access to a computer or account.

Simple real-life example: A password is like a key that allows an authorised user to enter an account.

Strong passwords

Book concept / teacher explanation: Use a mixture of uppercase letters, lowercase letters, numbers and special characters. Longer passwords are generally harder to guess.

Simple real-life example: Example: School@2026#Lab is an example for teaching only; never use a classroom example as your real password.

Avoid easy passwords

Book concept / teacher explanation: Do not use obvious information such as your name, birthday, 123456 or password.

Simple real-life example: Example: A password based only on your name may be easy for someone who knows you to guess.

Do not reuse passwords

Book concept / teacher explanation: Do not use exactly the same password for every important account.

Simple real-life example: Example: If the same password is used for email and another account and it is exposed, more than one account may be at risk.

2FA

Book concept / teacher explanation: Two-factor authentication adds another verification step after the password, such as a verification code.

Simple real-life example: Easy memory: Password = first lock; 2FA = second lock.

Phishing awareness

Book concept / teacher explanation: Do not enter passwords into suspicious websites or give them to people through suspicious messages.

Simple real-life example: Example: If a message asks you to urgently enter your password through an unknown link, verify it before taking action.

PAGE 112 – DATA BACKUP

Backup

Book concept / teacher explanation: A backup is an additional copy of important data.

Simple real-life example: Example: Keep another copy of an important school assignment so that it can be recovered if the computer fails.

Easy memory: Original data + Backup = safer data.

PAGE 112–113 – PHYSICAL SECURITY

Secure location

Book concept / teacher explanation: Keep computers in places where unauthorised people cannot easily access or steal them.

Simple real-life example: Example: A school computer laboratory can be locked after lessons and access can be controlled.

Cable locks

Book concept / teacher explanation: A cable lock can attach a computer to a fixed object, making it harder to remove or steal.

Simple real-life example: Example: A laptop in an office can be secured to a desk using a security cable.

Secure premises

Book concept / teacher explanation: Physical security may include locked doors, CCTV, alarms and access control systems.

Simple real-life example: Key point: Computer security is not only about passwords. The physical environment must also be protected.

PAGE 113 – CHAPTER SUMMARY

Main idea

Book concept / teacher explanation: Computer handling and care means protecting computer hardware, software, data and the environment where the computer operates.

Simple real-life example: Students should remember: Protect computers from liquid, dust, heat, unstable electricity, malware and physical damage.

Quick revision

Book concept / teacher explanation: Power: protect against brownouts, spikes and surges. Hardware: handle keyboards, mice and cables carefully. Software: use antivirus and updates. Data: make backups. Accounts: use strong passwords and 2FA. Physical security: prevent theft and unauthorised access.

Simple real-life example: Memory guide: UPS = backup power | Surge protector = surge protection | Stabiliser = stable voltage | Backup = extra copy | 2FA = second verification.

QUICK CLASSROOM QUESTIONS

1. What is a brownout?
2. What is a power spike?
3. What is a power surge?
4. What is the purpose of a UPS?
5. What is the difference between a UPS and a voltage stabiliser?
6. Why should liquids be kept away from computers?
7. Why should cables not be pulled by the wire?
8. Why is proper shutdown important?
9. What makes a password strong?
10. What is 2FA?
11. Why is data backup important?
12. What is physical security?
13. Give three ways of protecting a computer from theft.

ONE-LINE REVISION ANSWERS

Brownout	A temporary reduction in electrical voltage.
Spike	A sudden short increase in voltage.
Surge	A sudden significant increase in voltage.
UPS	A device that provides temporary backup power when the main power fails.
Surge protector	A device that helps protect equipment from voltage spikes and surges.
Voltage stabiliser	A device that helps maintain a more stable voltage supplied to equipment.
Antivirus	Software used to detect and protect against malicious software.
Backup	An additional copy of important data.
2FA	A security method requiring a second verification step in addition to a password.
Physical security	Measures used to protect computer equipment from theft, unauthorised physical access and physical damage.
Proper shutdown	Saving work, closing programs and using the operating system's shutdown procedure before powering off.